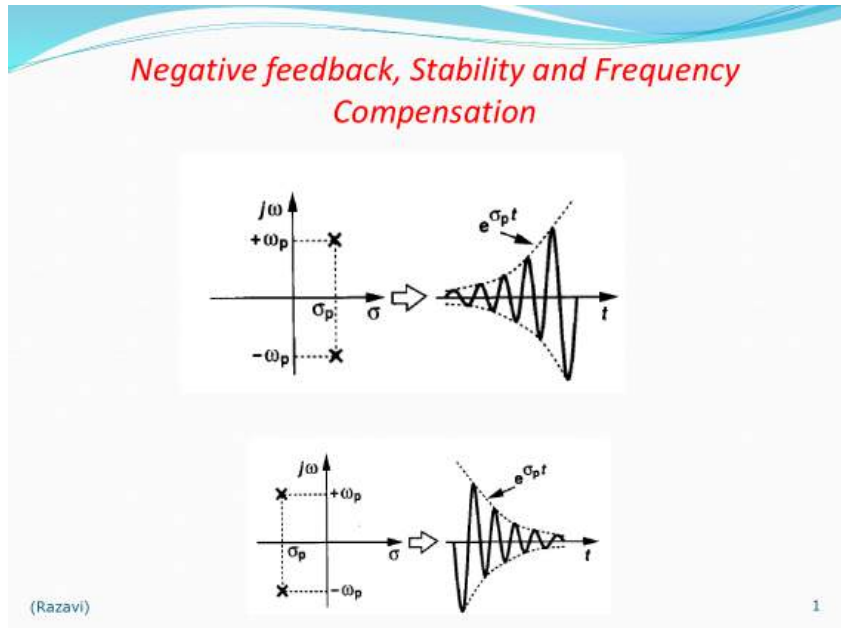


Lecture -26
Frequency Compensation

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→ Till now we studied about the Frequency response of many circuits starting from "Single Transistor" to "Folded Cascode Differential Amplifier".

We have studied particularly about

- Pole positioning and its impact on the Frequency Response

→ But in most applications our circuits will be in Feed back loops. So it is important to shape the Frequency Response of Feedback loops.

So, that the stability of the system can be ascertained.

→ Feedbacks are of Two types

- Negative
- positive

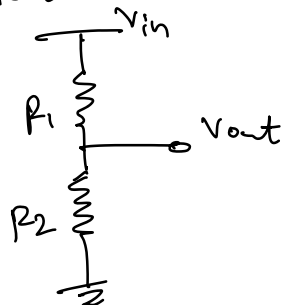
Negative Feedback

Motivation

- Certain applications, e.g., data converters, require amplifiers with a precise gain.
 - Relatively easy to make large imprecise gains.
 - Impractical to rely on r_o of device for precision.
- On the other hand, it's easy to make precise attenuators.
 - Can use ratio-metric designs (e.g., resistive dividers).

ex:- In Folded cascode diff. Amplifier we have studied it has a huge gain, but in many applications we would like to have a precise gain. So we make use of Negative Feedback.

→ If we want to implement an attenuation which is equal to $\frac{1}{2}$. The only thing we have to do is to create a Voltage Divider.

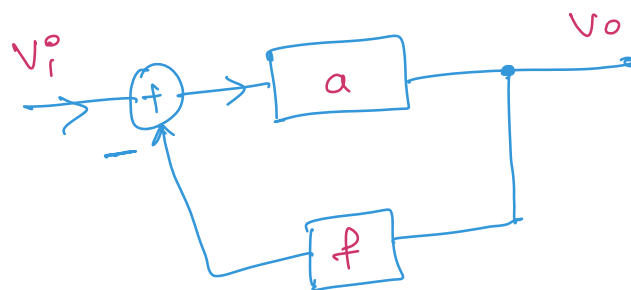


If $R_1 = R_2$

we can achieve very precisely $\frac{1}{2}$ of original input.

→ But there is no mean to implement a gain of for example 2 using Passive component.
and on the other hand Differential Amplifiers have very large but "Unprase" Gains.

∴ To control the gain of the circuit we use Negative feedback



Negative Feedback

$$v_o = a(v_i - f v_o) \Rightarrow \frac{v_o}{v_i} = \frac{a}{1 + af}$$

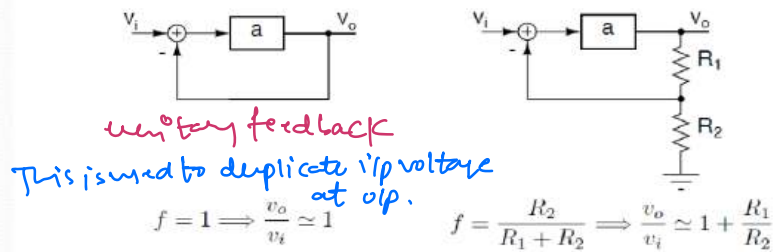
- Interesting case: $af \gg 1 \Rightarrow \frac{v_o}{v_i} \simeq \frac{1}{f}$
- As long as we maintain large gain in the forward path (a), the gain will depend only on f.

If 'a' is very large.

3

Precise Gain Using Precise Attenuation

- Since the gain is $1/f$, a small f is desirable.
 - e.g., stages of gain 2, popular in pipeline ADCs, require $f=0.5$.
- $f \leq 1$ is easy to implement, and can be ratiometric!
 - A "wire" ($f=1$) or resistive or capacitive voltage divider.



$$V_{P2} = \frac{V_o(P_2)}{R_1 + P_2} = \text{voltage across } P_2$$

$$\Rightarrow V_o = a(V_i - V_{P2})$$

$$V_o = a \left(V_i - \frac{V_o \cdot P_2}{R_1 + P_2} \right)$$

$$\Rightarrow V_o + a \frac{V_o P_2}{R_1 + P_2} = a V_i$$

$$\Rightarrow V_o \left(1 + \frac{a P_2}{R_1 + P_2} \right) = a V_i$$

$$\Rightarrow \frac{V_o}{V_i} = \frac{a}{1 + a \left(\frac{P_2}{R_1 + P_2} \right)} \rightarrow f$$



Example

$$R_1 = R_2 \implies f = 0.5$$

- Case I: $a=100$

$$\frac{v_o}{v_i} = \frac{a}{1 + af} = 1.96078 \dots$$

- Case II: $a=1000$

$$\frac{v_o}{v_i} = \frac{a}{1 + af} = 1.996007 \dots$$

- An order of magnitude change in forward gain causes less than 2% change in the closed-loop gain.

5

→ One fundamental ^{benefit} effect of this Feedback trick is also the fact that even a quite large variation of gain associated to open loop, generates a minor modification of the Global Gain of the circuit.

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Gain Desensitization

sensitivity of
closed loop gain

w.r to open loop gain.

$$A \triangleq \frac{v_o}{v_i} = \frac{a}{1+af}$$

$$\frac{dA}{da} = \frac{1}{(1+af)^2} \rightarrow \frac{\Delta A}{\Delta a} \simeq \frac{1}{(1+af)^2} \quad \star$$

$$\Rightarrow \frac{\Delta A}{A} \simeq \frac{\frac{\Delta a}{a}}{1 + \underbrace{af}_T}$$

- Fractional change in the forward gain (a) is reduced by af, which can be made arbitrarily large (at least in theory).
- The product of "a" and "f" is called the *loop gain* (T).
 - Sometimes also called the *loop transmission*.
- We will see shortly that T is a very meaningful parameter that will appear in bandwidth, stability and impedance calculations.

(Hekmat)

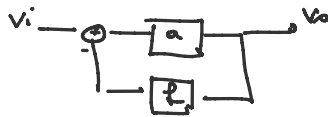
6



$$\frac{\Delta A}{A} = \frac{\Delta a}{(1+af)^2} \cdot \frac{1}{A}$$

$$\frac{\Delta A}{A} = \frac{\frac{\Delta a}{a}}{(1+af)^2} \cdot \frac{a}{1+af} = \frac{\Delta a}{a} \cdot \frac{1}{(1+af)^3}$$

$$\text{Nominal variation in } \textcircled{A} = \frac{\text{Nominal variation in } \textcircled{a}}{1+af}$$



Gain Desensitization

$$A \triangleq \frac{v_o}{v_i} = \frac{a}{1+af}$$

$$\frac{dA}{da} = \frac{1}{(1+af)^2} \rightarrow \frac{\Delta A}{\Delta a} \approx \frac{1}{(1+af)^2}$$

$$\Rightarrow \frac{\Delta A}{A} \approx \frac{\frac{\Delta a}{a}}{1+af}$$

- Fractional change in the forward gain (a) is reduced by af , which can be made arbitrarily large (at least in theory).
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- We will see shortly that T is a very meaningful parameter that will appear in bandwidth, stability and impedance calculations.

$$\frac{\Delta A}{A} = \frac{\frac{\Delta a}{a}}{(1+af)^2}$$

$$\frac{\Delta A}{A} = \frac{\frac{\Delta a}{a}}{(1+af)^2} = \frac{\Delta a}{a} \cdot \frac{1}{(1+af)^2}$$

Negative Feedback and Nonlinearity



- a_3 represents nonlinearity in transfer function.
 - 2nd order nonlinearity is assumed to be cancelled out using differential signaling.

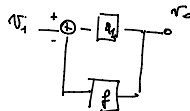
$$v_o = a_1(v_i - f v_o) + a_3(v_i - f v_o)^3$$

$$v_o = b_1 v_i + b_3 v_i^3$$

- Equating v_o equations and comparing coefficients we have:

$$b_1 = \frac{a_1}{1+a_1 f}, \quad b_3 = \frac{a_3}{(1+a_1 f)^4}$$

- The linear term is reduced by $(1+T)$.
- The cubic term is reduced by $(1+T)^4$.



$$\frac{v_o}{v_i} = \frac{a_1}{1+a_1 f} \rightarrow \frac{v_o}{a_1} = \frac{v_i - f v_o}{1}$$

$$\frac{v_o}{a_1} = \frac{v_i}{1+a_1 f}$$

$$\left(\frac{v_i}{1+a_1 f} \right) = (v_i - f v_o) \quad \text{①}$$

$$v_o = a_1 \underbrace{(v_i - f v_o)}_{\text{①}} + a_3 (v_i - f v_o)^3$$

$$v_o (1 + f a_1) = a_1 \cdot v_i + a_3 (v_i - f v_o)^3$$

$$v_o (1 + f a_1) = a_1 \cdot v_i + a_3 \cdot \frac{v_i^3}{(1 + a_1 f)^3}$$

$$v_o = \left[\frac{a_1}{(1 + f a_1)} \right] \cdot v_i + \left[\frac{a_3 \cdot v_i^3}{(1 + a_1 f)^4} \right]$$

$$\Downarrow$$

$$\triangleq b_1$$

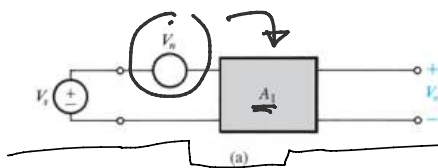
$$\triangleq b_3 \rightarrow 0$$



$$SNR_i = \frac{V_s}{V_n}$$

$$SNR_{out} = \frac{V_s \cdot A_2}{V_n}$$

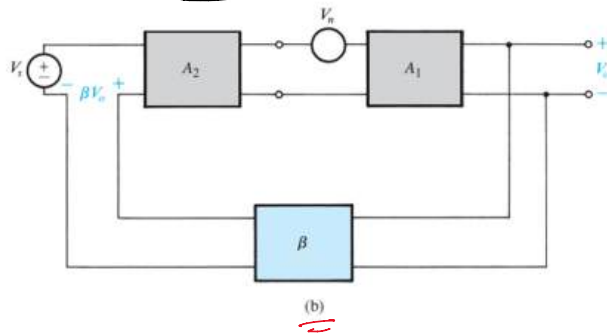
$$(b) \quad SNR = V_s \cdot A_2$$



$$SNR_i = \frac{V_s}{V_n}$$

$$SNR_{out} = \frac{V_s \cdot A_1}{V_n \cdot A_1}$$

$$(b) \quad SNR_{out} = \frac{V_s}{V_n} \cdot A_2$$

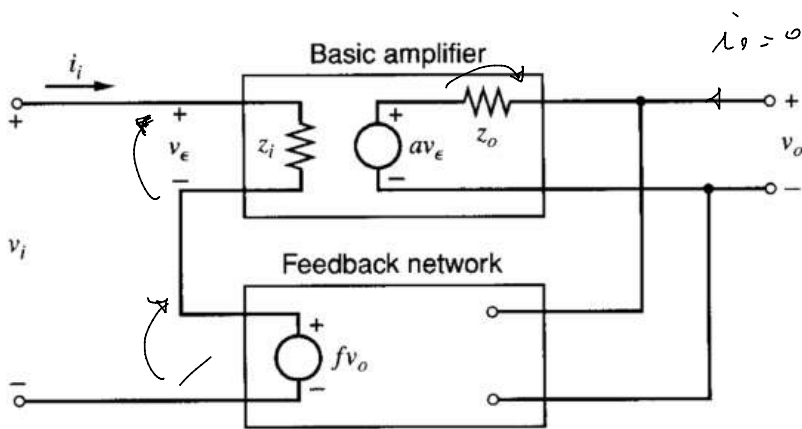


$$V_{os} = V_s \cdot \frac{A_2 \cdot A_1}{1 + \beta A_2 \cdot A_1}$$

$$V_{on} = V_n \cdot \frac{A_2}{1 + \beta A_2 \cdot A_1}$$

$$SNR_{out} = \frac{V_{os}}{V_{on}} = \frac{V_s \cdot A_2 \cdot A_1}{V_n \cdot A_1} = \frac{V_s}{V_n} \cdot A_2$$

$$f = \beta A_2$$



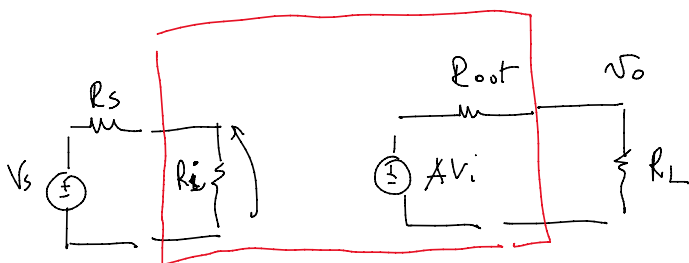
$$v_o = v_e \cdot a$$

$$v_i = v_e + f \cdot v_o$$

$$v_i = v_e + f \cdot a \cdot v_e$$

$$v_i = v_e (1 + f \cdot a)$$

$$i_i = \frac{v_e}{Z_i}$$



$$R_i \gg R_s$$

$$R_{out} \ll R_L$$

$$\frac{v_o}{i_o} = Z_o = \frac{v_e (1 + f \cdot a)}{\frac{v_e}{Z_i}} = Z_i (1 + f \cdot a)$$

$$\frac{v_o}{i_o} = \frac{Z_o}{(1 + a f)} = Z_o$$

$$\frac{v_o}{i_o} \bigg|_{v_i=0}$$

$$v_o = a v_e + i_o \cdot Z_o$$

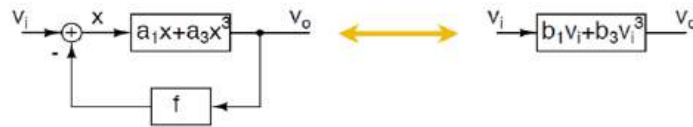
$$\frac{v_o - a v_e}{Z_o} = i_o$$

$$v_e + f v_o = 0$$

$$v_{\epsilon} = -\frac{p}{f} v_o$$

$$\frac{v_o + \frac{p}{f} v_o}{Z_o} = i_o$$

Negative Feedback and Nonlinearity



- a_3 represents nonlinearity in transfer function.
 - 2nd-order nonlinearity is assumed to be cancelled out using differential signaling.

$$v_o = a_1(v_i - f v_o) + a_3(v_i - f v_o)^3$$

$$v_o = b_1 v_i + b_3 v_i^3$$

- Equating v_o equations and comparing coefficients we have:

$$b_1 = \frac{a_1}{1 + a_1 f}, \quad b_3 = \frac{a_3}{(1 + a_1 f)^4}$$

- The linear term is reduced by $(1+T)$.
- The cubic term is reduced by $(1+T)^4$.

→ This means we are diminishing Non-linearity using feedback

(Hekmat)

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→ In most of our applications we are assuming i/p-o/p relation is truly linear.

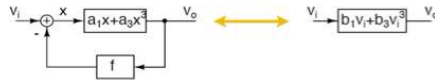
If we have Resistance, Capacitance & Inductances then circuit is truly linear.

→ But if we have Non-linear operation we linearize it. This is typically the case when we analyze MOSFET & BJT using SSM and find the i/p-o/p relationship.



How do we handle Non-linearity and what role does Negative Feedback play?

Negative Feedback and Nonlinearity



- a_3 represents nonlinearity in transfer function.
- 2nd order nonlinearity is assumed to be cancelled out using differential signaling.

$$v_o = a_1(v_i - f v_o) + a_3(v_i - f v_o)^3 \quad \checkmark$$

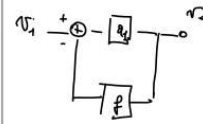
$$v_o = b_1 v_i + b_3 v_i^3$$

- Equating v_o equations and comparing coefficients we have:

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- The linear term is reduced by $(1+T)$.
- The cubic term is reduced by $(1+T)^4$.

(Hekmat)



$$\frac{v_o}{v_i} = \frac{a_1}{1 + a_1 f} \rightarrow \frac{v_o}{a_1} = \frac{v_i - f v_o}{1}$$

$$\frac{v_o}{a_1} = \frac{v_i}{1 + a_1 f} \quad \text{--- (2)}$$

$$\left(\frac{v_i}{1 + a_1 f} \right) = (v_i - f v_o) \quad \text{--- (1)}$$

$$\textcircled{1} = \textcircled{2}$$

$$v_o = a_1 (v_i - f v_o) + a_3 (v_i - f v_o)^3$$

both are equivalent

$$v_o (1 + f a_1) = a_1 \cdot v_i + a_3 (v_i - f v_o)^3$$

$$v_o (1 + f a_1) = a_1 \cdot v_i + a_3 \cdot \frac{v_i^3}{(1 + a_1 f)^3}$$

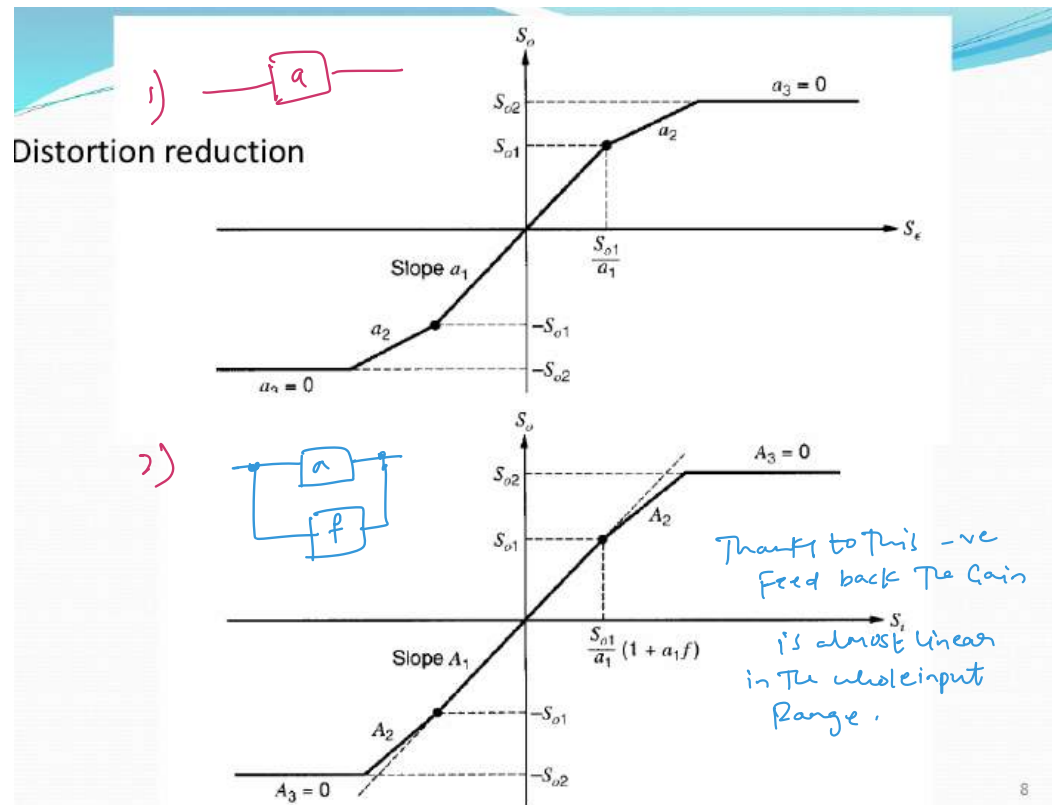
$$v_o = \left[\frac{a_1}{(1 + f a_1)} \right] \cdot v_i + \left[\frac{a_3 \cdot v_i^3}{(1 + a_1 f)^4} \right]$$

$$\Downarrow$$

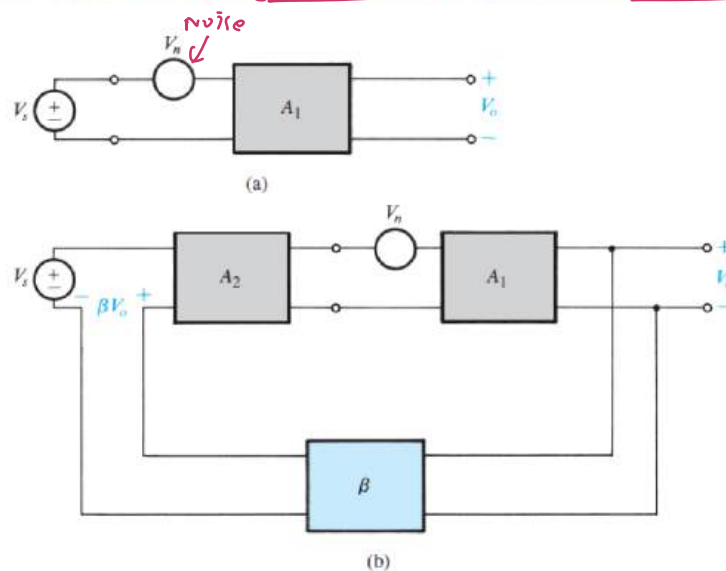
$$\triangleq b_1$$

$$\triangleq b_3 \rightarrow 0$$

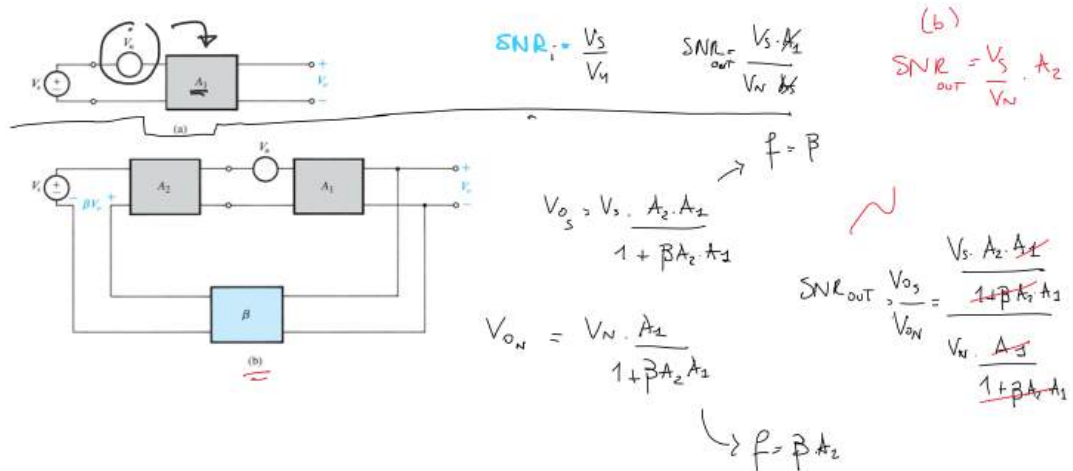
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EBIT, Unib O



Minimize the interferences' impact



→ Here we are going to study the impact of Noise on the circuit operation and how does Feedback help us in reducing the Noise Interference in the Global circuit behaviour.

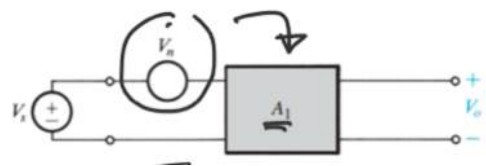


→ we want to verify if we plug the noise at the input the openloop circuit is much more deformed wrt to the interference being in b/w the feedback path

to accentuate benefit we can compute signal to noise ratio (SNR)

$$SNR_i = \frac{V_s}{V_n}$$

$$SNR_o = \frac{A_1 V_s}{V_n}$$



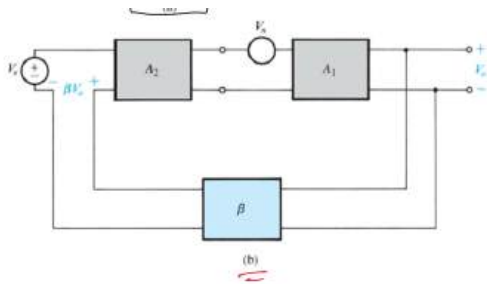
$$A_1 = A_2 \text{ in}$$

$$A_s \cdot V_n$$

open loop

$$\therefore \boxed{\text{SNR}_0 = \frac{V_s}{V_n}} \text{ for open loop}$$

Signal to Noise Ratio in closed loop



$$V_{o_s} = V_s \cdot \frac{A_2 \cdot A_1}{1 + \beta A_2 A_1}$$

$$f = \beta$$

$$V_{o_n} = V_n \cdot \frac{A_2}{1 + \beta A_2 A_1}$$

$$f = \beta A_2$$

$$\text{SNR}_{\text{OUT}} = \frac{V_{o_s}}{V_{o_n}} = \frac{V_s \cdot \frac{A_2 \cdot A_1}{1 + \beta A_2 A_1}}{V_n \cdot \frac{A_2}{1 + \beta A_2 A_1}}$$

First let's look at V_{o_s} (Vout wrt to source)

$$\therefore \boxed{V_{o_s} = V_s \frac{A_2 \cdot A_1}{1 + \beta A_2 A_1}} \quad \text{--- (1)}$$

op generated by the Noise according to configuration

$$\boxed{V_{o_n} = V_n \frac{A_1}{1 + \beta A_1 A_2}} \quad \text{Important Remember} \quad \text{--- (2)}$$

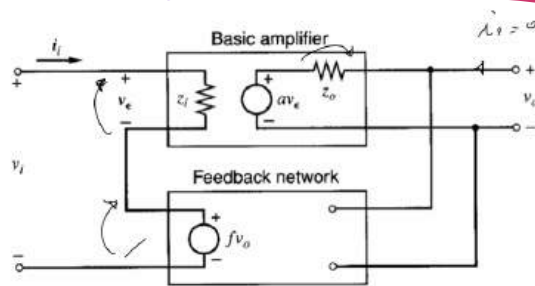
Now using (1) + (2) we can compute signal to Noise Ratio.

$$\text{SNR}_{\text{out}} = \frac{V_{os}}{V_{on}} = \frac{V_s A_2}{V_n}$$

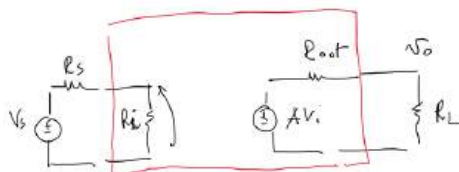
(for feedback loop)

* Another important aspect of using Feedback is its effect on I_p & O_p Impedances

For voltage gain we want High i_p Impedence & low o_p Impedence



$$\begin{aligned} V_o &= A_v V_e \\ V_i &= V_e + f \cdot V_o \\ V_i &= V_e + f \cdot A_v \cdot V_e \\ V_i &= V_e (1 + f A_v) \\ i_i &= \frac{V_e}{Z_i} \end{aligned}$$



$$\frac{V_i}{i_i} = Z_i = \frac{V_e (1 + f A_v)}{\frac{V_e}{Z_i}} = Z_i (1 + f A_v)$$

$$R_i \gg R_s$$

$$R_{out} \ll R_L$$

This condition should be satisfied to transfer as much as information to the load.

$$\frac{V_o}{i_o} = \frac{Z_o}{(1 + a f)} = Z_o$$

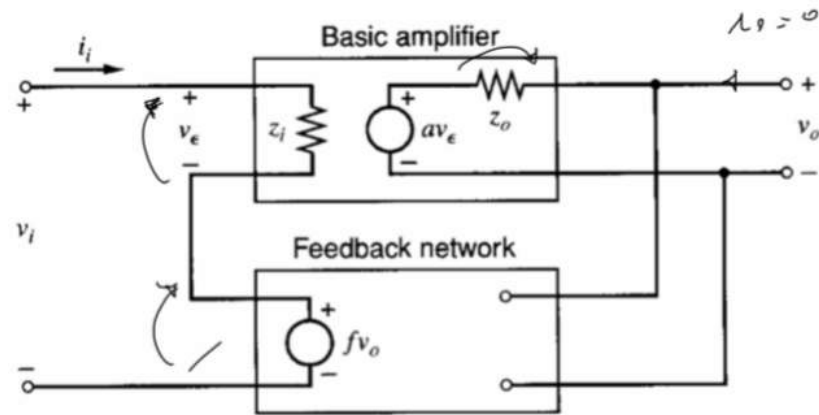
$$\left. \frac{V_o}{i_o} \right|_{V_e = 0} = Z_o$$

$$V_e + f V_o = 0$$

$$\begin{aligned} V_o &= A_v V_e + i_o \cdot Z_o \\ \frac{V_o - A_v V_e}{Z_o} &= i_o \end{aligned}$$

$$V_e = -f V_o$$

$$\frac{V_o + A_v f V_o}{Z_o} = i_o$$



→ Input Impedance:
First calculating Input Impedance by considering an open circuit at the o/p

⇒ $i_o = 0$, $V_o = aV_e$ *
(∵ No current is flowing through Z_o)

→ The i/p voltage on the other side is given by combination of voltage across Z_i and feedback voltage fv_o

$$V_i = V_e + fv_o$$

$$\Rightarrow V_i = V_e + faV_e$$

$$V_i \Rightarrow V_e (1 + fa)$$

$$I_i = \frac{V_e}{Z_i} \quad \text{Actual I/n Impedance}$$

$$z_i \rightarrow \dots$$

$$\left(\frac{V_i}{I_i} \right)_{I_o=0} = \text{Input Impedance} = Z_i \rightarrow \text{Impedance because of feedback}$$

$$\Rightarrow Z_i = \frac{V_i}{I_i} \Rightarrow \frac{V_e(1+fa)}{V_e/Z_i}$$

$$Z_i \Rightarrow (1+fa) Z_i$$

Total I/p Impedance because of feedback
Actual i/p impedance

* output Impedance

$$\left. \frac{V_o}{I_o} \right|_{V_i=0} =$$

$$\Rightarrow V_i = V_e + f V_o = 0$$

$$V_e = -f V_o$$

$$\Rightarrow V_o = a V_e + i_o z_o$$

$$\frac{V_o - aV_e}{Z_o} = i_o$$

$$\Rightarrow \frac{V_o - a(-fV_o)}{Z_o} = i_o$$

$$\Rightarrow \frac{V_o + afV_o}{Z_o} = i_o$$

$$\Rightarrow \frac{V_o(1+af)}{Z_o} = i_o$$

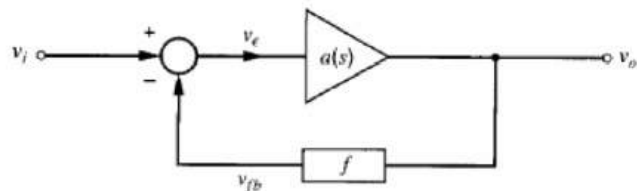
$$Z_o \Rightarrow \boxed{\frac{V_o}{i_o} = \frac{Z_o}{1+af}}$$

∴ Impedance is Reduced

* Another interesting Aspect of -ve Feedback is increase in Band width *

Band width in Negative Feedback

Negative Feedback and Bandwidth



$$a(s) = \frac{a_0}{1 - \frac{s}{p_1}}$$

$$A(s) = \frac{v_o}{v_i} = \frac{a(s)}{1 + a(s)f} = \frac{a_0}{1 + a_0 f} \frac{1}{1 - \frac{s}{p_1} \frac{1}{1 + a_0 f}}$$

$$A_0 = \frac{a_0}{1 + T_0}$$

$$T_0 = a_0 f = \text{low-frequency loop gain}$$

→ Here you can see the pole is multiplied by a factor $(1 + a_0 f)$. So this means that angular frequency associated with pole is moved to much higher frequency.

i.e Bandwidth Increases

→ The interesting aspect is ^{that} original original DC gain is reduce i.e a_0 is DC gain

$$\Rightarrow \frac{a_0}{1 + a_0 f} \rightarrow \text{New Gain}$$

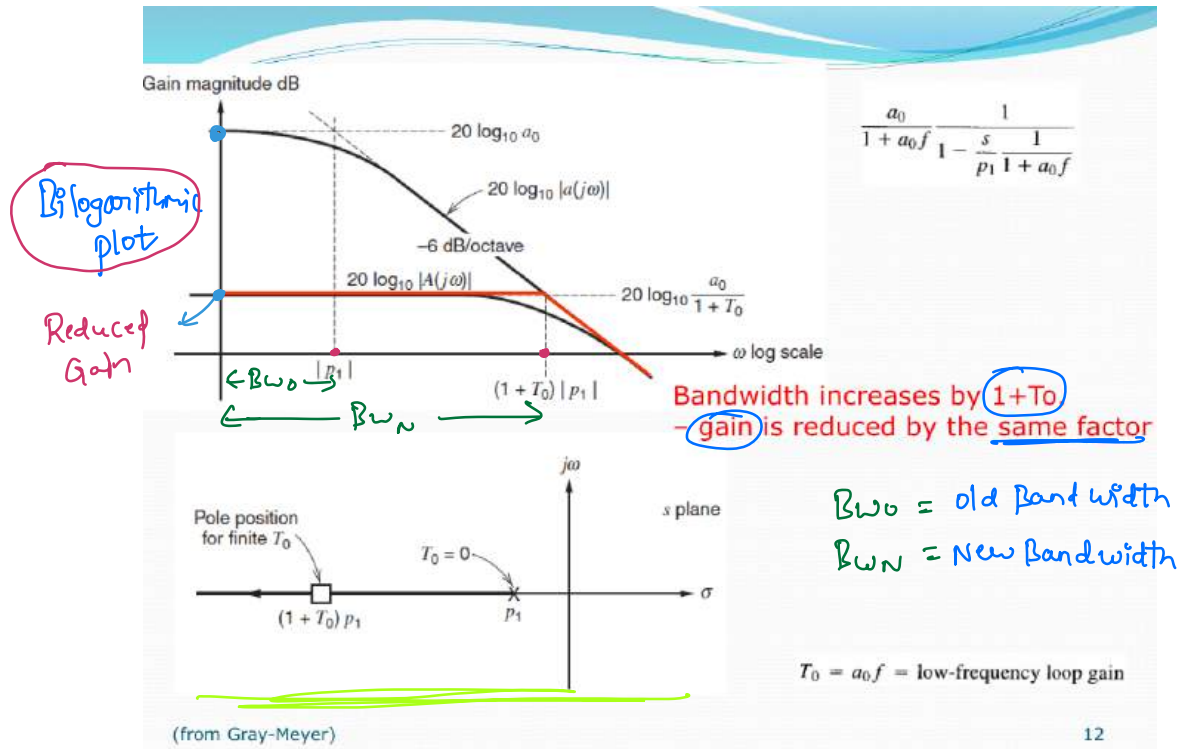
Basically we are lowering Gain and increasing Bandwidth.

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effect of -ve feedback shown in plot

(from Gray-Meyer textbook)

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Benefits of negative feedback

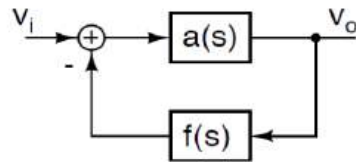
- Precise gain, gain desensitization, non linearity reduction
- Interferences desensitization
- Optimization of input and output impedances
- Bandwidth enlargement

Stability?

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Q What about the stability of the system?

Stability



$$A(s) = \frac{v_o}{v_i} = \frac{a(s)}{1 + a(s)f(s)} = \frac{a(s)}{1 + T(s)}$$

The most general criterion for stability:

- **B**ounded **I**nterface - **B**ounded **O**utput (BIBO).
- Applicable to any system.
- A continuous time linear system is BIBO stable if all its poles are in the left half of the s-plane:
- Can calculate roots of $[1+T(s)]$ to check stability.
- Tedious and hard to do in general, but...
- We will look in detail at $T(s)$...

(Hekmat)

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Nyquist Criterion

- A feedback system is unstable when $|T(j\omega)| > 1$ at the frequency where $\text{Phase}[T(j\omega)] = -180^\circ$

- 1) Phase margin

- Defined at the frequency where $|T(j\omega)| = 1 \rightarrow \omega_0$

$$\text{PM} = 180^\circ + \text{Phase}[T(j\omega_0)]$$

- 2) Gain margin

- Defined at the frequency where $\text{Phase}[T(j\omega)] = -180^\circ \rightarrow \omega_{180}$

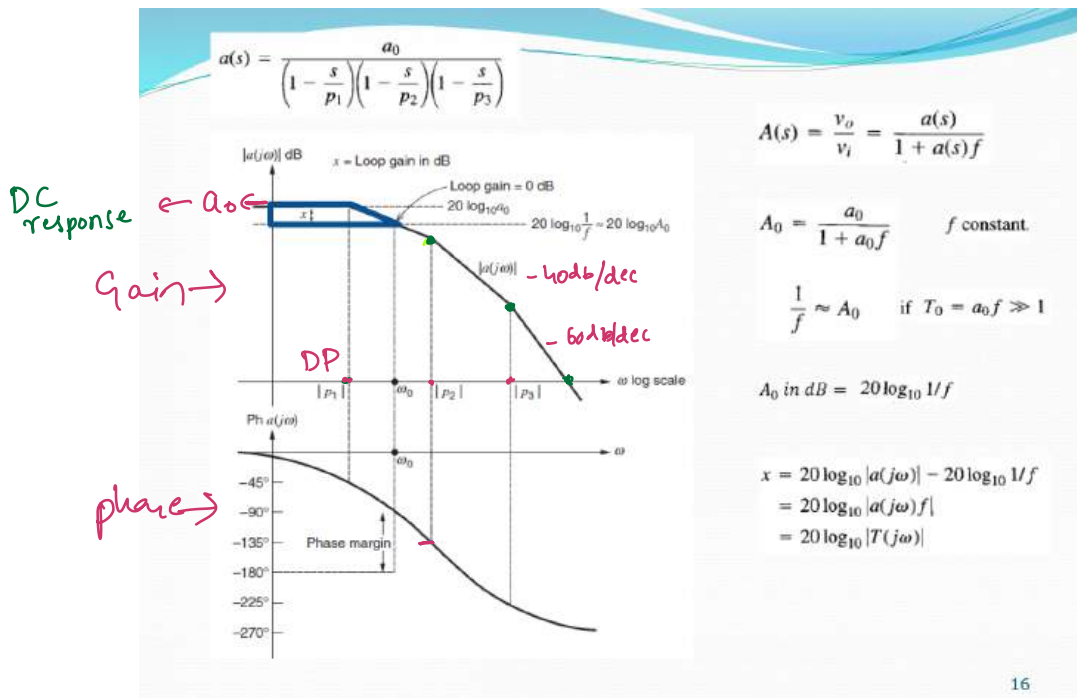
$$\text{GM} = \frac{1}{|T(j\omega_{180})|}$$

If this GM is greater than unity we have some margin otherwise system is unstable.

Typical PM should be larger than 35° or 45° or even 60°

usually this quantity is used to check stability

* let's investigate to determine phase margin
this plot



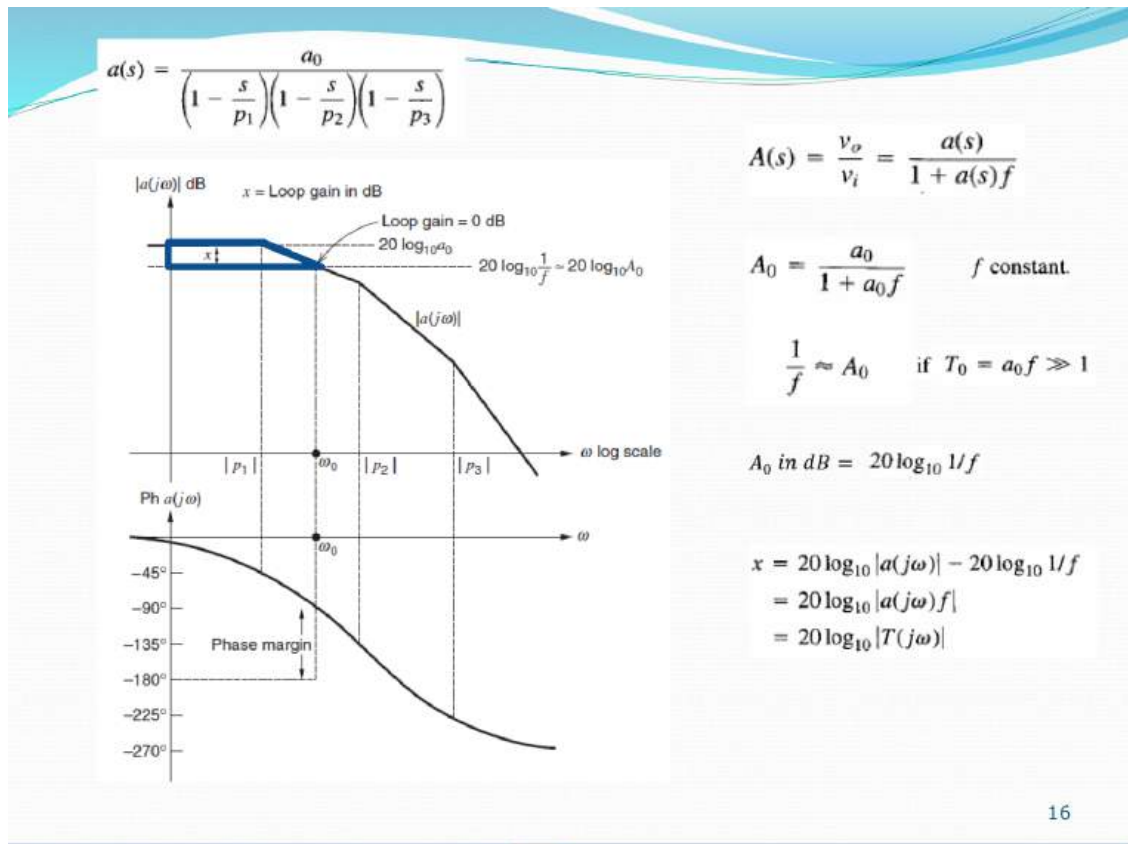
we know a pole causes a Gain loss of -20 dB/dec
and phase loss of -90°

True 90° is lost in two parts
Half of 90° i.e. 45° is lost before
and another 45° after P_1 in about 1 dec
after this pole.

But what we are interested is in $a f$
i.e. $T(f) = a f$

✓ ✓ ✓

✱



→ Let's investigate this plot to determine phase margin in this specific case. when Transfer function is as above.

$$a(s) = \frac{a_0}{\left(1 - \frac{s}{p_1}\right)\left(1 - \frac{s}{p_2}\right)\left(1 - \frac{s}{p_3}\right)}$$

→ Here there are 3 poles in the Transfer function

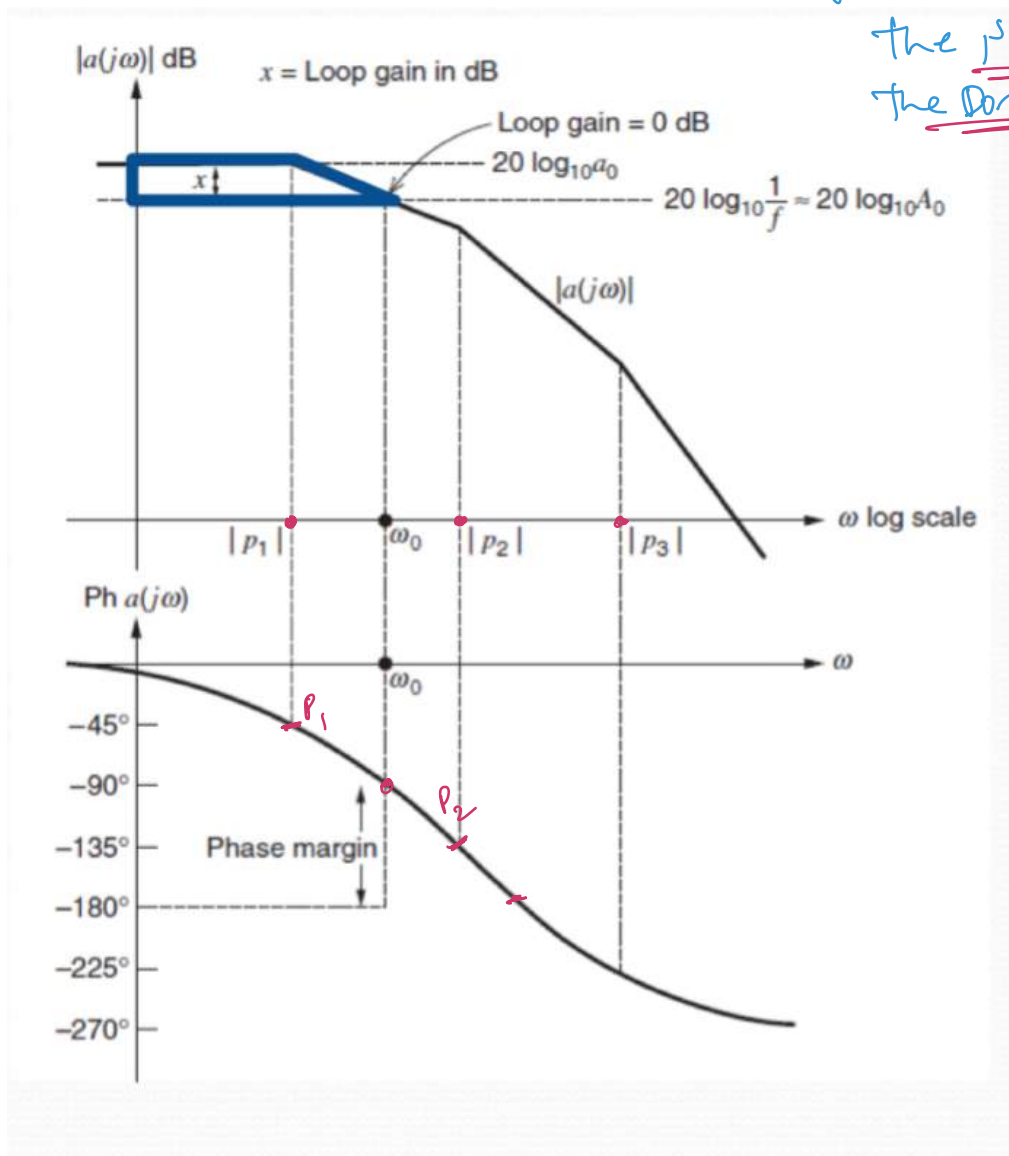
$$p_1, p_2, p_3$$

→ whenever we have poles we can construct Bodeplot.

Firstly, $\boxed{\text{DC Response} = a_0}$

$\therefore$ we start from a_0 , then we get to an angular frequency of

the 1st Pole $\textcircled{P_1}$
the Dominant
one



$\rightarrow$ After reaching first pole $\textcircled{P_1}$ we will start to lose -20dB/dec, after certain point we will also find point $\textcircled{P_2}$ and from $\textcircled{P_3}$ we will

Start losing -40 dB/dec

Finally, we will get to Angular frequency corresponding to P_3 and from this point on we will be losing -60 dB/sec .

Q What about The Phase Margin?

So, you may remember that each pole will cause a phase loss equal to 90°

At P_1 , 90° are lost in two parts,

Half of the 90° is 45° is lost before pole position and the other half will be lost after 1 dec after P_1

Continuing, from P_1 we will start losing 45° w.r to P_2 .

So when we get to P_2 , The phase loss is equal to 135° , immediately 1 dec after P_2 remaining 45° will be lost and total

phase loss after p_2 will be 180° and

as we continue further towards p_3 more phase loss will happen.



$$a(s) = \frac{a_0}{\left(1 - \frac{s}{p_1}\right)\left(1 - \frac{s}{p_2}\right)\left(1 - \frac{s}{p_3}\right)}$$

This is the transfer function, $a(s)$ intensity of frequency response.

But, what we are interested is determining the effect of af

This is the quantity we have to check to infer stability according to Nyquist Criteria

$$\therefore T = af$$

Where is the position, when $af = 1$ in the pole plot?

$$T = af = 1 \Rightarrow \underbrace{P_h(T) - (-180^\circ)}_{\text{phase margin}}$$

✓

*** phaseMargin = $\text{ph}(T(j\omega_{dB}) - (-180^\circ))$

PM = $\text{ph}(T(j\omega_{dB}) + 180^\circ)$

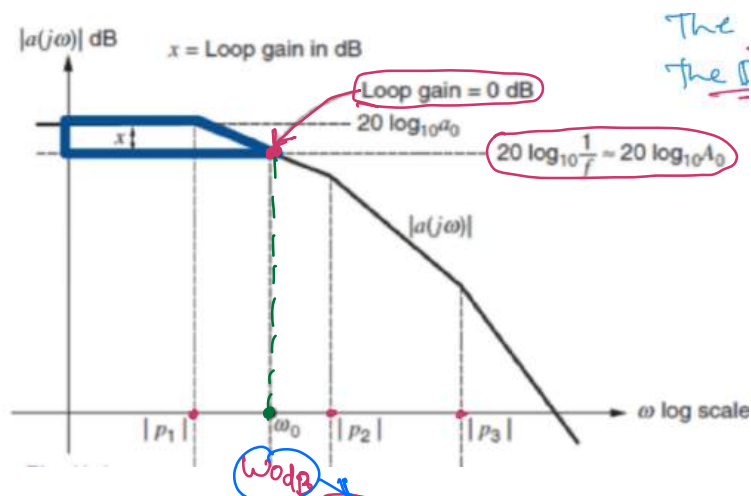
$T(j\omega_{dB}) = 1 = a(\omega_{dB}) \cdot f$

$\Rightarrow a(\omega_{dB}) = \frac{1}{f}$

$A(s) = \frac{v_o}{v_i} = \frac{a(s)}{1 + a(s)f}$
 $A_0 = \frac{a_0}{1 + a_0 f} \quad f \text{ constant.}$
 $\frac{1}{f} \approx A_0 \quad \text{if } T_0 = a_0 f \gg 1$
 $A_0 \text{ in dB} = 20 \log_{10} 1/f$

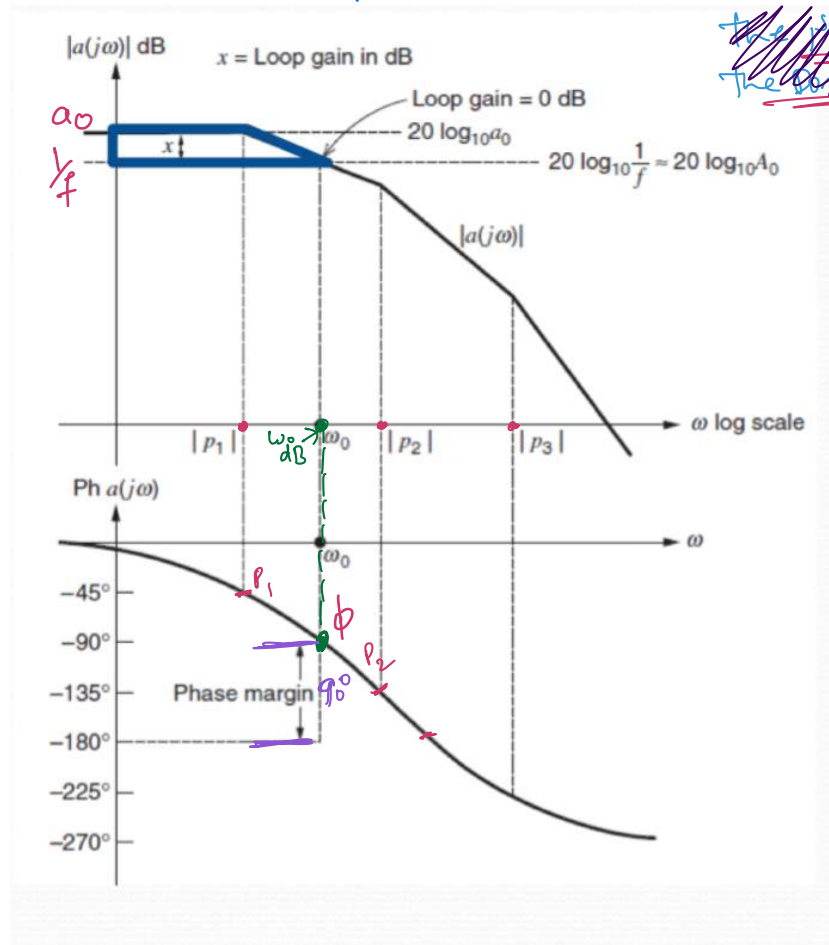
we can transform this expression in decibels by

$20 \log_{10} a(\omega_{dB}) = 20 \log_{10} A_0 = 20 \log_{10} \frac{1}{f}$
 at frequency ω_0



when $a = 1 \rightarrow$ we will get this position

→ So ω_0 is the one which we use to compute the phase margin



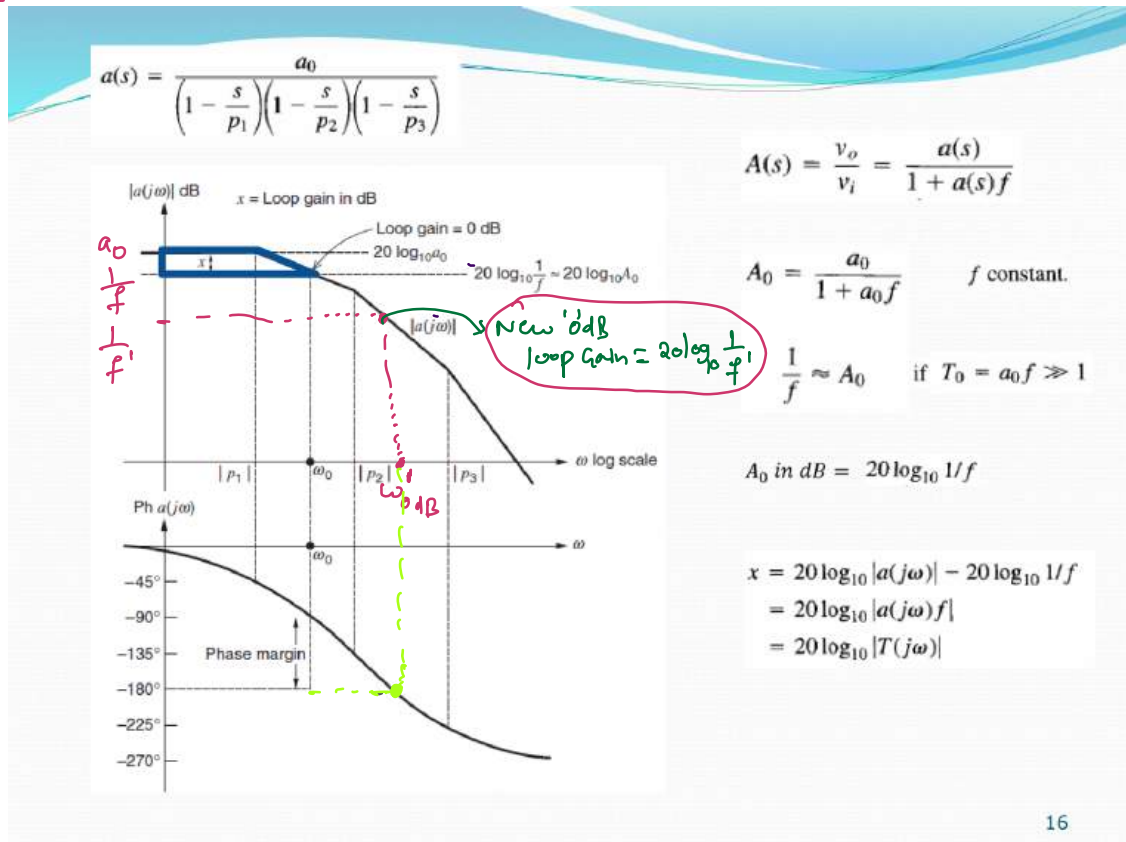
→ ϕ is the corresponding phase of ω_{dB} in the above example the phase margin is

90° because

$$\begin{aligned} \text{PM} &= 180^\circ + \text{phase}(T(j\omega_{dB})) \\ &= 180^\circ - 90^\circ = 90^\circ \end{aligned}$$

Note:-

Here we have quite large Phase Margin so this system is stable.



→ In this case, we may realize that $\frac{1}{f}$ is relatively High value.

→ as we know that $f < 1$ because it is realized with passive components like Resistors.

→ So, we can assume f is very very small

4 $\frac{1}{f}$ is high and as a consequence of that the Angular Frequency which corresponds to 0dB loop gain is quite is quite small. It is just after P_1

* $\rightarrow$ If we increase f slightly to f' then the value of $\frac{1}{f} = \frac{1}{f'}$ decreases and move the 0dB gain to lower values

For this new 0dB gain the corresponding the corresponding new ω_{0dB} will be large and "The phase margin will be almost negligible as we have seen in the Bode plot above."

$\rightarrow$ So, we have certain dependency on ' f ' to determine stability.

Q What is the value of f for which the stability will be ~~poor~~?

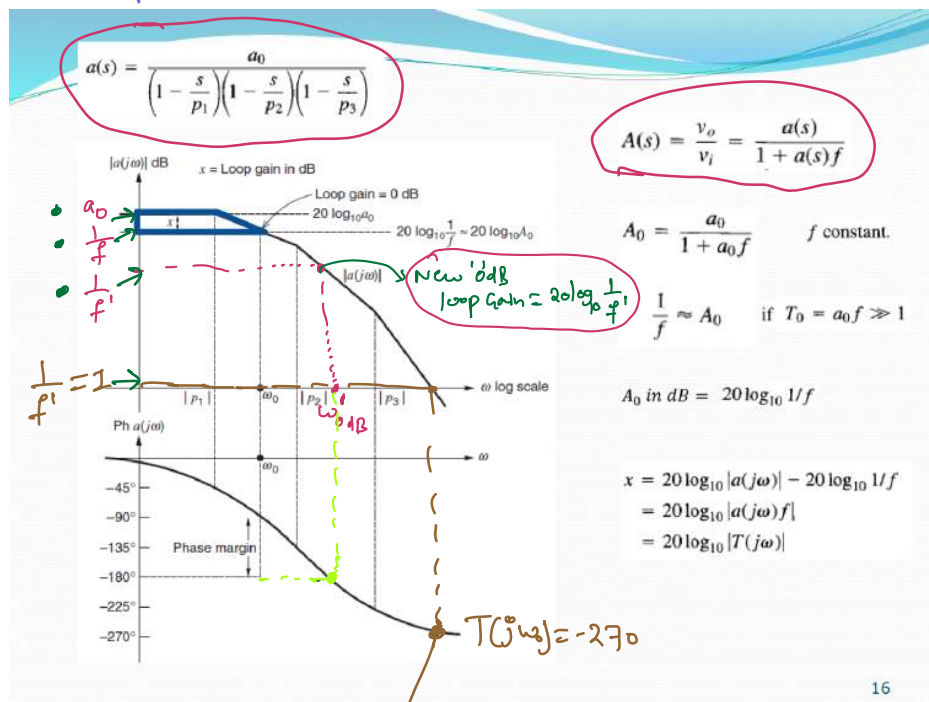
Ans

$$\boxed{f = 1}$$

Then

$$\text{0 dB gain} = 20 \log_{10} \left(\frac{1}{f} \right) = 20 \log_{10}(1) = \underline{\underline{0}}$$

→ Then the 0 dB ^{loop} gain line will be on the frequency axis and will touch the Bode plot after P_3 and the corresponding phase margin will be -ve which represents Unstability.



The corresponding PM

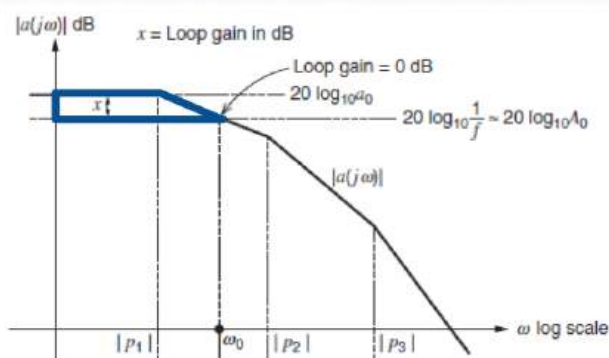
$$PM = -270 + 180$$

$$= -90$$

Unstable System.



Loop gain magnitude



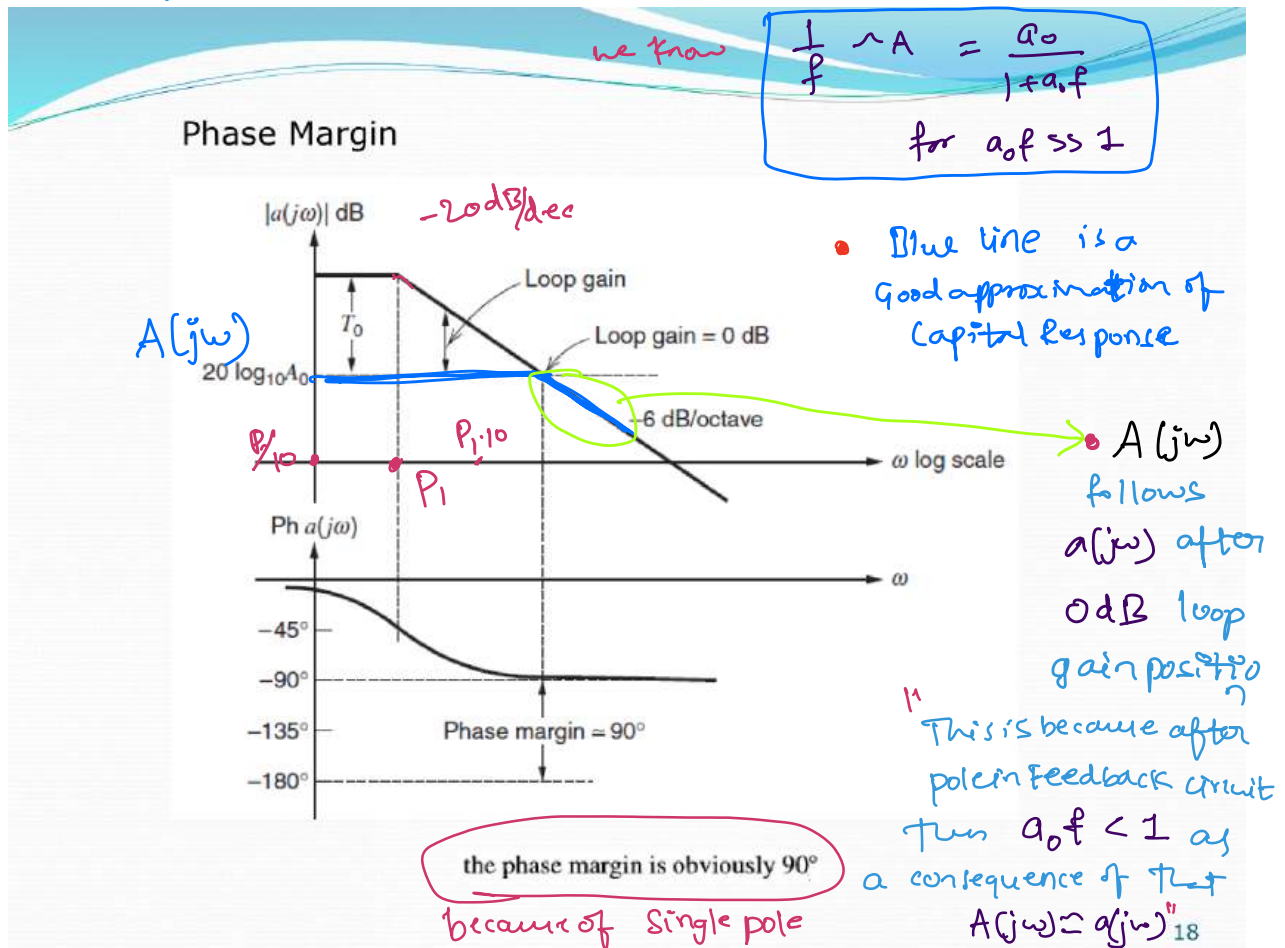
$$\begin{aligned} x &= 20 \log_{10} |a(j\omega)| - 20 \log_{10} 1/f \\ &= 20 \log_{10} |a(j\omega)f| \\ &= 20 \log_{10} |T(j\omega)| \end{aligned}$$

Thus the distance x is a *direct measure in decibels* of the loop-gain magnitude, $|T(j\omega)|$. The point where the curve of $20 \log_{10} |a(j\omega)|$ intersects the line $20 \log_{10} 1/f$ is the point where the loop-gain magnitude $|T(j\omega)|$ is 0 dB or unity, and the curve of $|a(j\omega)|$ in decibels can thus be considered a curve of $|T(j\omega)|$ in decibels if the dotted line at $20 \log_{10} 1/f$ is taken as the new zero axis.



We will now consider several conditions for Phase Margin to better understand

What could be the appearance of Frequency Response after the generation of these Feedback loops.

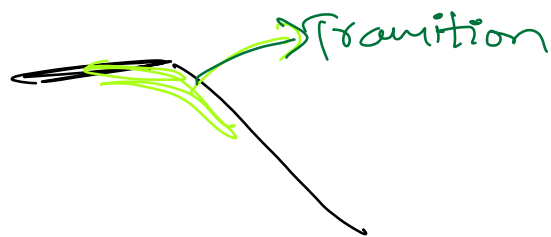


→ In the above Graph we can see it a situation with single pole. These single pole once again generates -45° loss before pole position and -45° after 1 dec after pole position. So in total 90° phase is lost.

→ The vast majority of phase loss occurs between $P_1/10$ & $P_1 \cdot 10$

→ The pole position also acts on magnitude curve and after P_1 there is -20dB/dec loss.

→ Please, remember that this is just asymptotic representation of actual Frequency Response, you may be familiar with the concept that we have smoother transition & this



→ we have already have -3dB loss in the amplitude of the curve.

Whenever we have a single pole the stability

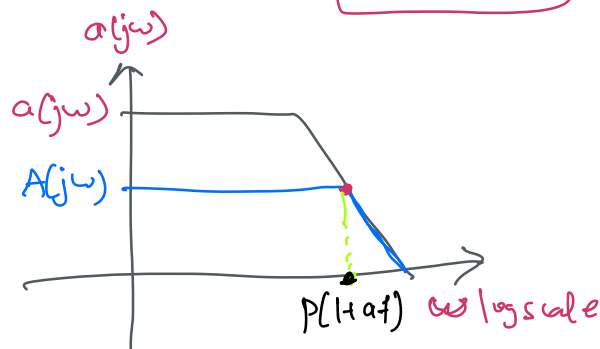
issues generally do not arise. what
 ever $\frac{1}{f}$ value we pick up still the
 phase margin will be around 90° .
 because in any circumstances ω_{0dB}
 will be in correspondence to phase
 margin i.e. 90°

$$\rightarrow 20 \log_{10} A_0 = 20 \log_{10} \frac{1}{f} = 0dB \text{ loop Gain}$$

But this is also a very good
 approximation of Gain of whole circuit.

$$A_0 = \frac{a_0}{1 + a_0 f} \quad \text{if } a_0 f \gg 1$$

$$\text{then } A_0 \approx \frac{1}{f}$$



• $\rightarrow a(jw)$
 • $\rightarrow A(jw)$



After pole in Feedback circuit

$$a_0 f \ll 1$$

$$\text{Then } A(j\omega) \approx a(j\omega)$$

This is true when Angular frequency is much larger than $p_1(1+a_0f)$

In this condition we have smooth transition from Flatband part to decaying part.

Consider a feedback amplifier with a phase margin of 45°

PM = 45°

$$\text{ph } T(j\omega_0) = -135^\circ$$

where ω_0 is the frequency defined by

$$\text{open loop transfer function } |T(j\omega_0)| = 1$$

Now $|T(j\omega_0)| = |a(j\omega_0)f| = 1$ implies that

$$\text{assuming that } f \text{ is positive real. } |a(j\omega_0)| = \frac{1}{f}$$

The overall gain is

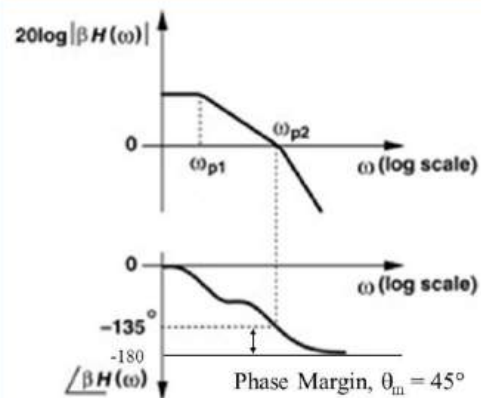
$$A(j\omega) = \frac{a(j\omega)}{1 + T(j\omega)}$$

$$A(j\omega_0) = \frac{a(j\omega_0)}{1 + e^{-j135^\circ}} = \frac{a(j\omega_0)}{1 - 0.7 - 0.7j} = \frac{a(j\omega_0)}{0.3 - 0.7j}$$

$$T(j\omega_0) = e^{-j135^\circ} \quad \text{we know } |T(j\omega)| = 1$$

$$|A(j\omega_0)| = \frac{|a(j\omega_0)|}{0.76} = \frac{1.3}{f}$$

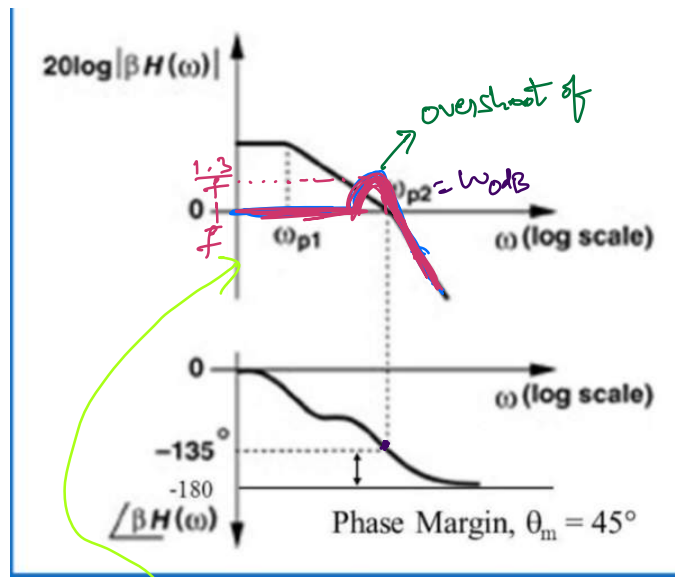
The frequency ω_0 , where $|T(j\omega_0)| = 1$, is the nominal -3 -dB point for a single-pole basic amplifier, but in this case there is 2.4 dB ($1.3 \times$) of peaking above the low-frequency gain of $1/f$.



$$|A(j\omega_0)| = \frac{|a(j\omega_0)|}{0.76} = \frac{1.3}{f}$$

It would be interesting to note that, because this means, when we have this condition

here $|A(j\omega_0)| = \frac{1.3}{f} = \frac{|a(j\omega_0)|}{0.76}$



$|A(j\omega_0)|$ curve over shoots at $|a(j\omega_0)| = \frac{1.3}{f}$ close to second pole in the graph which is coincident with ω_{0dB}

*** The overshoot peaks at value $a(j\omega_0) = \frac{1.3}{f}$ and then the graph of $A(j\omega_0)$ follows

Note:- $a(j\omega_0)$

This is the effect of phase margin = 45°

It generates this overshoot. If you repeat this same analysis for phase margin = 60° for the same circuit.

✱ Consider a phase margin of 60° . At the frequency ω_0 in this case $PM=60^\circ$

$$\text{ph } T(j\omega_0) = -120^\circ$$

Following a similar analysis we obtain

$$|A(j\omega_0)| = \frac{1}{f}$$

Here $\omega_{0dB} < \omega_{2nd\text{ pole}}$

In this case there is no peaking at $\omega = \omega_0$, but there has also been no gain reduction at this frequency.

✱ Finally, the case where the phase margin is 90° can be similarly calculated. In this case $PM=90^\circ$

$$\text{ph } T(j\omega_0) = -90^\circ$$

A similar analysis gives

$$|A(j\omega_0)| = \frac{0.7}{f}$$

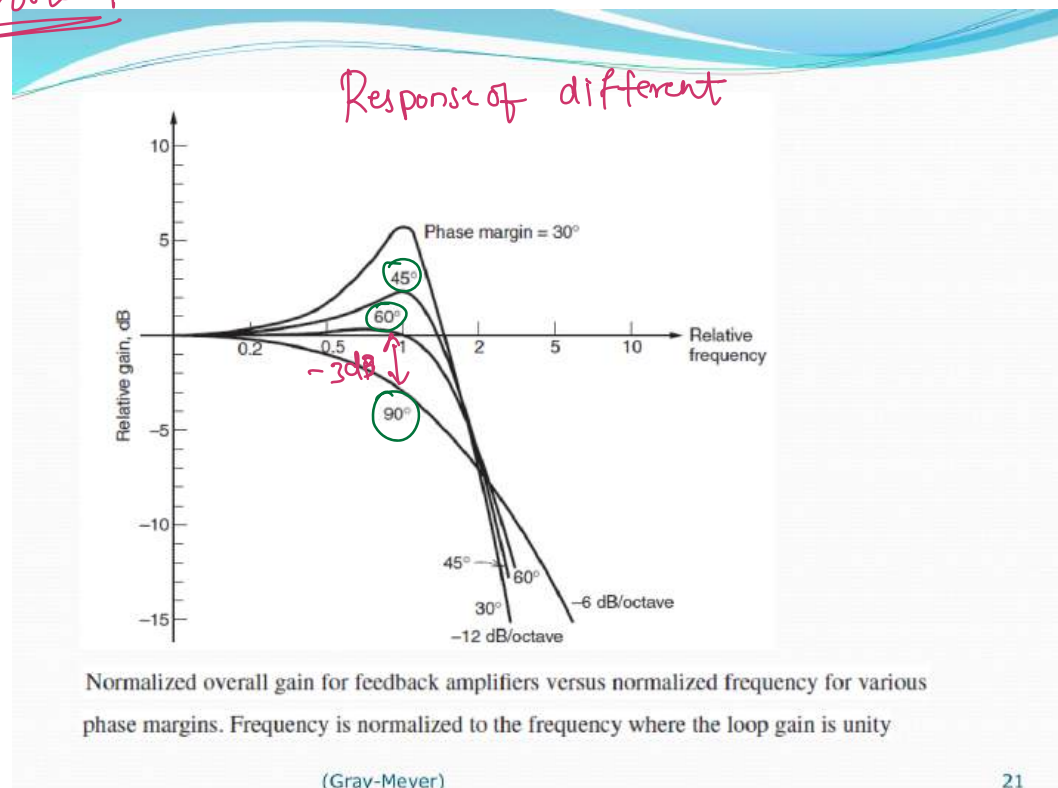
This is the 1st condition we analyzed above

As expected in this case, the gain at frequency ω_0 is 3 dB below the midband value.

GM

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Important

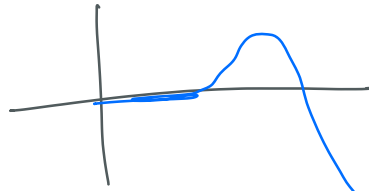


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→ The point here is that the phase margin has link to frequency response of overall circuit. This link is mostly related to overshoot that we may have depending on actual phase margin value.

So, the frequency response will be the one of the low pass filter. We will have certain

Flat region and will have overshoot.



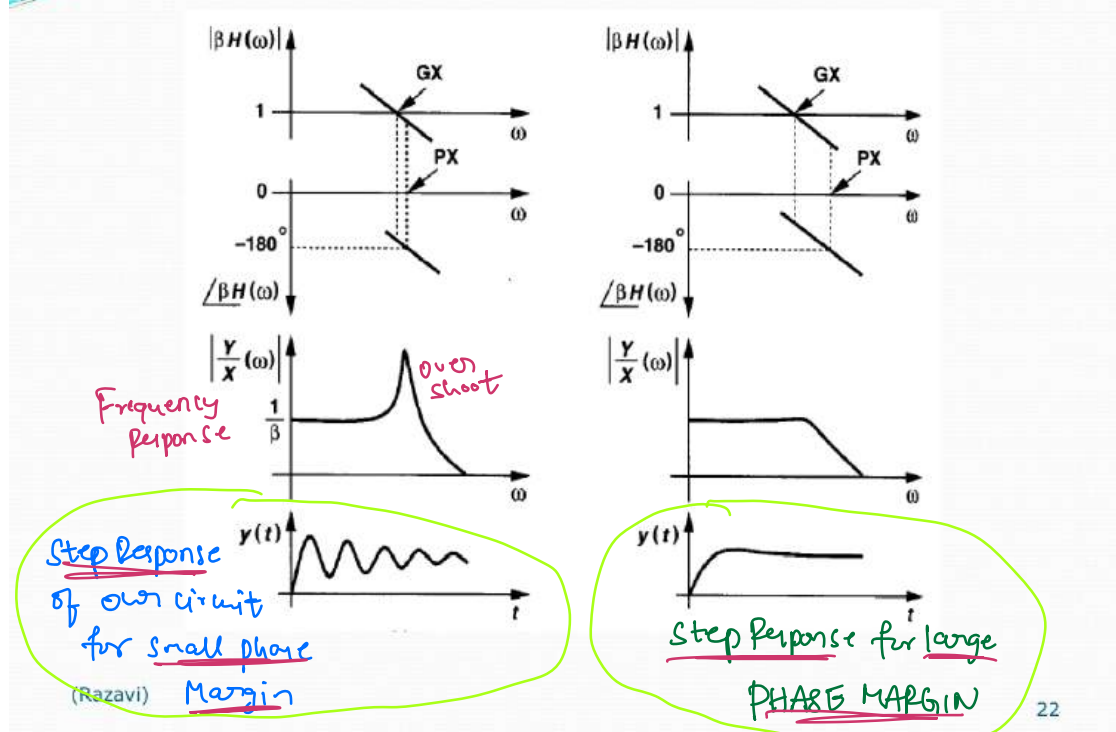
and smaller is the phase margin, the higher is the overshoot.

→ If $PM=0$ we will have basically instability that is the reason we have very high overshoot.



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Phase Margin and Gain Margin effects can also be seen also in terms of Step Response effects.



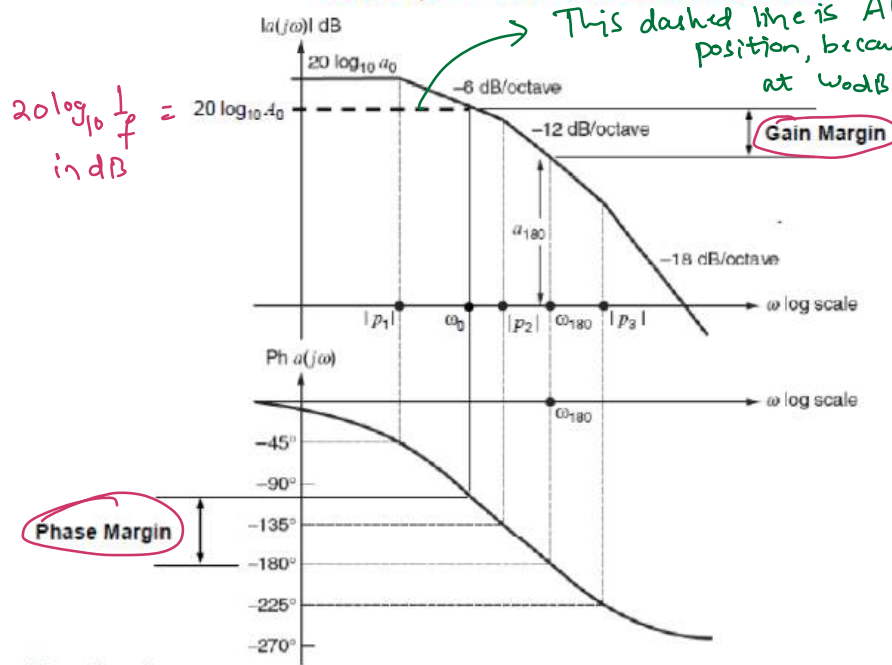
This is the Qualitative view of Phase Margin effect.

→ But we can use the below analysis to study PM and ensure stability.



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Example: the Circuit is Stable



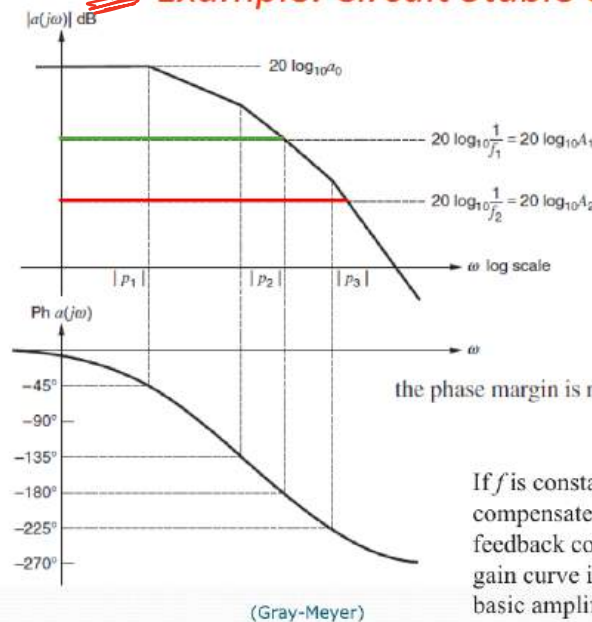
This dashed line is $A(j\omega)$ in safe position, because if we look at ω_{0dB} we can see

that, phase curve is consistently apart from -180° curve so the system is STABLE

23

8

Example: Circuit Stable or Not?



$$f = f_1$$

OK

$$f = f_2$$

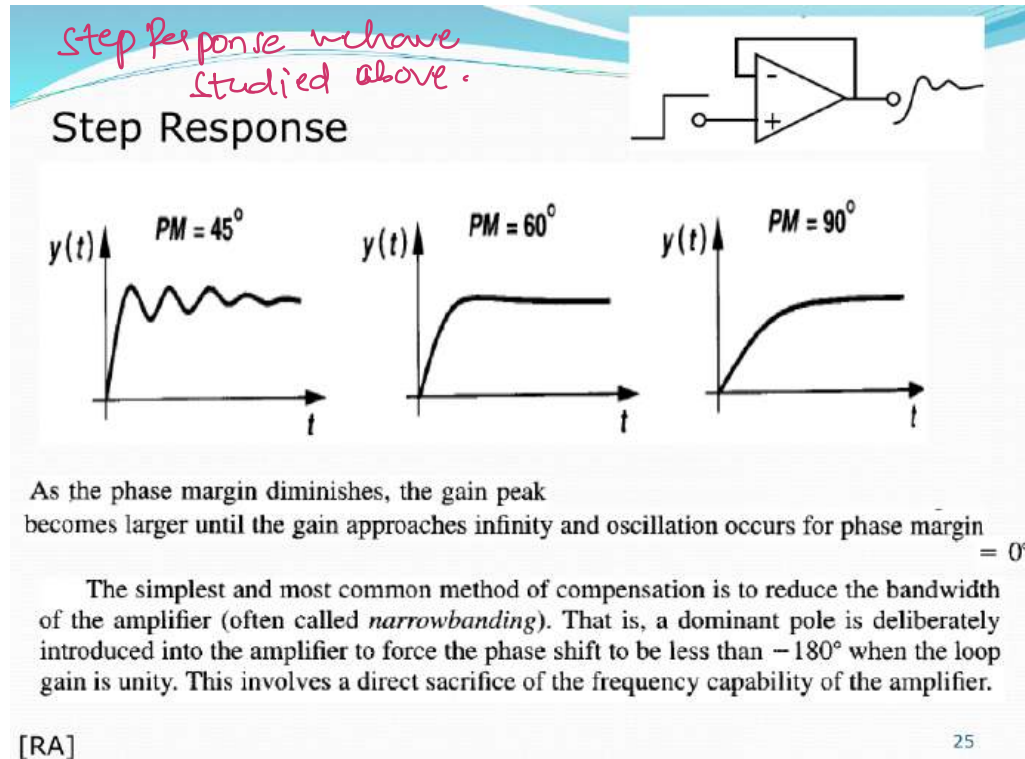
KO

If the feedback is increased to f_2 the phase margin is negative and the circuit will oscillate.

If f is constant, the most difficult case to compensate is $f = 1$, which is a unity-gain feedback configuration. In this case the loop-gain curve is identical to the gain curve of the basic amplifier ($T(s) = a(s) \cdot f = a(s)$)

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The Graph is self explanatory.



How can we enforce stability in a circuit with lower Phase Margin?

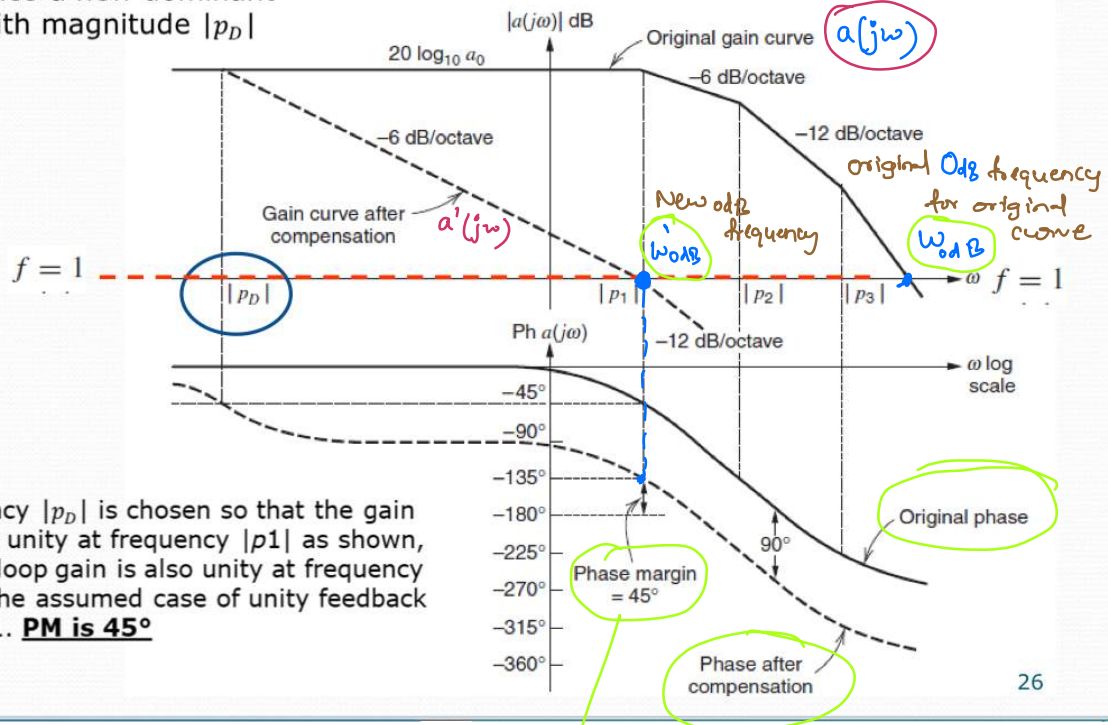
To purposely add an additional pole.

This is called Frequency Compensation with Dominant pole.



Frequency Compensation with Dominant Pole (Narrowbanding)

Introduce a new dominant pole with magnitude $|p_D|$



If frequency $|p_D|$ is chosen so that the gain $|a(j\omega)|$ is unity at frequency $|p_1|$ as shown, then the loop gain is also unity at frequency $|p_1|$ for the assumed case of unity feedback with $f = 1$. **PM is 45°**

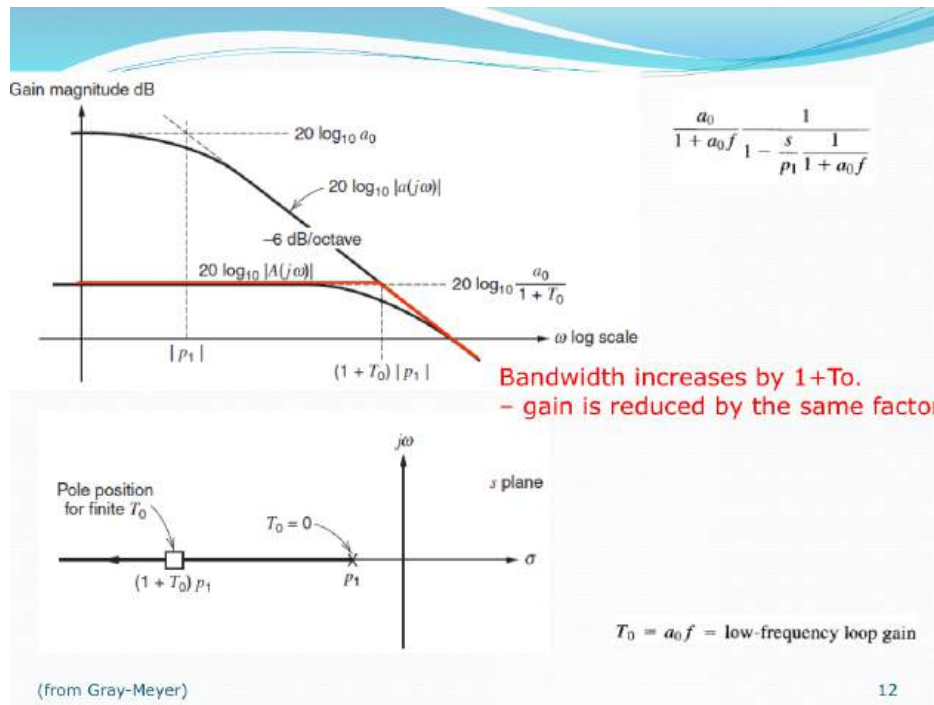
~~*~~ New phase margin after compensation even we are in the worst possible condition with feedback $f=1$

This situation was achieved by

$$\omega_{0dB} = p_1$$

Q

How do we achieve above condition by choosing a new P_D ?



The answer lies in the analysis.

Basically

$$P_1 = a_0 \cdot P_D$$

Q

Dominant poles are introduced by adding capacitors in our amplifier?

- You may remember that Dominant poles

→ are related to Time constant of our circuit from earlier class.

→ so, we add additional capacitance to a given node to the ground to add this dominant pole in the circuit.

→ There are also many alternatives which we will study in the next class.

By

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